

MARIADB

TASKS

1.Install MariaDB on your Linux system and verify the installation by running the command to check the MariaDB server version.

Install MariaDB and Verify

Step 1: Install MariaDB

`dnf install mariadb-server -y`

Step 2: Start

```
root@localhost:~  
Verifying      : mariadb-server-3:10.5.29-4.el9.x86_64      8/13  
Verifying      : mariadb-server-utils-3:10.5.29-4.el9.x86_64 9/13  
Verifying      : mysql-selinux-1.0.15-1.el9.noarch          10/13  
Verifying      : perl-DBD-MariaDB-1.21-17.el9.x86_64        11/13  
Verifying      : perl-File-Copy-2.34-483.el9.noarch          12/13  
Verifying      : perl-Sys-Hostname-1.23-483.el9.x86_64      13/13  
  
Installed:  
mariadb-3:10.5.29-4.el9.x86_64  
mariadb-backup-3:10.5.29-4.el9.x86_64  
mariadb-common-3:10.5.29-4.el9.x86_64  
mariadb-connector-c-3.2.6-1.el9.x86_64  
mariadb-connector-c-config-3.2.6-1.el9.noarch  
mariadb-errmsg-3:10.5.29-4.el9.x86_64  
mariadb-gssapi-server-3:10.5.29-4.el9.x86_64  
mariadb-server-3:10.5.29-4.el9.x86_64  
mariadb-server-utils-3:10.5.29-4.el9.x86_64  
mysql-selinux-1.0.15-1.el9.noarch  
perl-DBD-MariaDB-1.21-17.el9.x86_64  
perl-File-Copy-2.34-483.el9.noarch  
perl-Sys-Hostname-1.23-483.el9.x86_64  
  
Complete!  
[root@localhost ~]#
```

MariaDB

`systemctl start mariadb`

Step 3: Enable MariaDB

`systemctl enable mariadb`

Step 4: Check Status

`systemctl status mariadb`

Step 5:

```
root@localhost:~ — systemctl status mariadb
• mariadb.service - MariaDB 10.5 database server
  Loaded: loaded (/usr/lib/systemd/system/mariadb.service; enabled; preset: >
  Active: active (running) since Thu 2026-07-16 00:14:51 IST; 33s ago
    Docs: man:mariabdd(8)
          https://mariadb.com/kb/en/library/systemd/
  Main PID: 56132 (mariabdd)
    Status: "Taking your SQL requests now..."
     Tasks: 11 (limit: 4272)
  Memory: 87.5M (peak: 117.6M)
     CPU: 853ms
  CGroup: /system.slice/mariadb.service
          └─56132 /usr/libexec/mariabdd --basedir=/usr

Jul 16 00:14:50 localhost.localdomain mariadb-prepare-db-dir[56089]: The second>
Jul 16 00:14:50 localhost.localdomain mariadb-prepare-db-dir[56089]: you need t>
Jul 16 00:14:50 localhost.localdomain mariadb-prepare-db-dir[56089]: After conn>
Jul 16 00:14:50 localhost.localdomain mariadb-prepare-db-dir[56089]: able to co>
Jul 16 00:14:50 localhost.localdomain mariadb-prepare-db-dir[56089]: See the Ma>
Jul 16 00:14:50 localhost.localdomain mariadb-prepare-db-dir[56089]: Please rep>
Jul 16 00:14:50 localhost.localdomain mariadb-prepare-db-dir[56089]: The latest>
Jul 16 00:14:50 localhost.localdomain mariadb-prepare-db-dir[56089]: Consider j>
Jul 16 00:14:50 localhost.localdomain mariadb-prepare-db-dir[56089]: https://ma>
Jul 16 00:14:51 localhost.localdomain systemd[1]: Started MariaDB 10.5 database>
lines 1-23
```

Verify Version
mariadb --version

or
mysql --version

```
[root@localhost ~]# mariadb --version
mariadb Ver 15.1 Distrib 10.5.29-MariaDB, for Linux (x86_64) using EditLine wr
apper
[root@localhost ~]# mysql --version
mysql Ver 15.1 Distrib 10.5.29-MariaDB, for Linux (x86_64) using EditLine wrap
per
[root@localhost ~]#
```

2. Create a new MariaDB database named foodie_app and a table restaurants with columns id (INT, primary key), name (VARCHAR), cuisine (VARCHAR), and rating (FLOAT). Insert 3 sample restaurant records.

Create Database, Table and Insert Records

Step 1: Login

mariadb

Prompt changes to:
MariaDB [(none)]>

```
[root@localhost ~]# mariadb
Welcome to the MariaDB monitor.  Commands end with ; or \g.
Your MariaDB connection id is 3
Server version: 10.5.29-MariaDB MariaDB Server

Copyright (c) 2000, 2018, Oracle, MariaDB Corporation Ab and others.

Type 'help;' or '\h' for help. Type '\c' to clear the current input statement.

MariaDB [(none)]>
```

Step 2: Create Database

```
CREATE DATABASE foodie_app;
```

Step 3: Use Database

```
USE foodie_app;
```

Step 4: Create Table

```
CREATE TABLE restaurants (
id INT PRIMARY KEY,
name VARCHAR(100),
cuisine VARCHAR(50),
rating FLOAT
);
```

Step 5: Insert Records

```
INSERT INTO restaurants VALUES
(1,'Pizza Hut','Italian',4.2),
(2,'Dominos','Italian',3.9),
(3,'Spice Villa','Indian',4.8);
```

Step 6: Verify

```
SELECT * FROM restaurants;
```

```
MariaDB [foodie_app]> SELECT * FROM restaurants;
+-----+-----+-----+-----+
| id | name      | cuisine | rating |
+-----+-----+-----+-----+
| 1  | Pizza Hut | Italian | 4.2    |
| 2  | Dominos  | Italian | 3.9    |
| 3  | Spice Villa | Indian | 4.8    |
+-----+-----+-----+-----+
3 rows in set (0.001 sec)

MariaDB [foodie_app]>
```

3. Write and execute a SQL query in MariaDB to find all restaurants in the foodie_app database with a rating greater than 4.0 and display their name and cuisine.

Find Restaurants Rating Greater Than 4

Run

```
SELECT name,cuisine
FROM restaurants
WHERE rating > 4.0;
```

```
MariaDB [foodie_app]> SELECT name,cuisine
-> FROM restaurants
-> WHERE rating > 4.0;
+-----+-----+
| name      | cuisine |
+-----+-----+
| Pizza Hut  | Italian |
| Spice Villa | Indian  |
+-----+-----+
2 rows in set (0.001 sec)

MariaDB [foodie_app]>
```

4. Grant SELECT and INSERT permissions on the foodie_app database to a new MariaDB user named zomato_user with password zomato123, then test the permissions by connecting as that user and inserting a new restaurant.

Create User and Grant Permissions

Step 1: Create User

```
CREATE USER 'zomato_user'@'localhost'
IDENTIFIED BY 'zomato123';
```

Step 2: Grant Permissions

```
GRANT SELECT, INSERT
ON foodie_app.*
TO 'zomato_user'@'localhost';
```

Step 3: Apply Changes

```
FLUSH PRIVILEGES;
```

Step 4: Exit

```
EXIT;
```

Step 5: Login as New User

```
mariadb -u zomato_user -p
```

Password:
zomato123

Step 6: Select Database

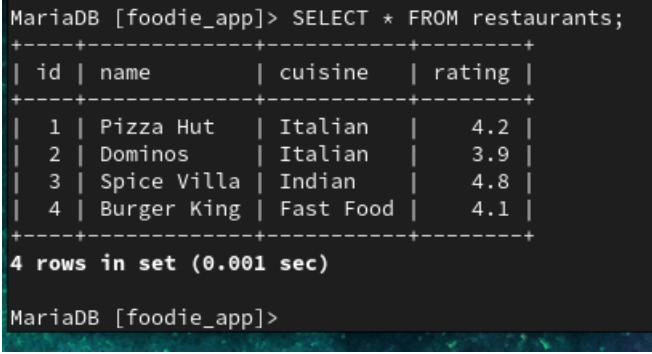
USE foodie_app;

Step 7: Insert Record

```
INSERT INTO restaurants  
VALUES  
(4,'Burger King','Fast Food',4.1);
```

Step 8: Verify

```
SELECT * FROM restaurants;
```



```
MariaDB [foodie_app]> SELECT * FROM restaurants;  
+-----+-----+-----+-----+  
| id | name      | cuisine | rating |  
+-----+-----+-----+-----+  
| 1  | Pizza Hut | Italian | 4.2    |  
| 2  | Dominos  | Italian | 3.9    |  
| 3  | Spice Villa | Indian | 4.8    |  
| 4  | Burger King | Fast Food | 4.1    |  
+-----+-----+-----+-----+  
4 rows in set (0.001 sec)  
  
MariaDB [foodie_app]>
```

5. Use ChatGPT or Copilot to generate a SQL query that updates the rating of all restaurants with cuisine 'Italian' to 4.5 in your MariaDB table, then run the query and verify the update.
Hint: Paste your prompt and the AI-generated query in your submission along with the result.

Update Italian Restaurants Rating

Prompt Used

Write a SQL query to update the rating of all restaurants with cuisine 'Italian' to 4.5.

AI Generated Query

```
UPDATE restaurants  
SET rating = 4.5  
WHERE cuisine='Italian';
```

Verify :

```
SELECT * FROM restaurants;
```

```
MariaDB [foodie_app]> SELECT * FROM restaurants;
```

id	name	cuisine	rating
1	Pizza Hut	Italian	4.5
2	Dominos	Italian	4.5
3	Spice Villa	Indian	4.8
4	Burger King	Fast Food	4.1

```
4 rows in set (0.001 sec)
```

```
MariaDB [foodie_app]> 
```